Bazzite AI Smart Home

Extension Guide — EX.0 through EX.6

AI chat · Voice assistant · Home automation · Nextcloud file access

Hardware: GMKtec NucBox K8 · Ryzen 7 8845HS · Radeon 780M (gfx1103) · 32 GB DDR5

Requires: Bazzite Unified Home Server + Gaming Station guide completed — all Phase 0–11 containers healthy.

What this guide adds: Ollama GPU inference · Open WebUI · mcpo (filesystem + memory tools) · Home Assistant · Wyoming voice pipeline (Whisper STT + Piper TTS) · full remote access via HTTPS. No new hardware required for EX.0–EX.6.

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ROCM + 780M — DEFINITIVE ASSESSMENT

The Radeon 780M (gfx1103 / RDNA3) is usable for GPU inference on Bazzite (Fedora 42). Three environment variables and one kernel parameter are mandatory. Without them, Ollama falls back to CPU or the iGPU suffers an unrecoverable hang under sustained inference.

Required Kernel Parameter

> **CRITICAL** Apply this BEFORE any other step. Reboot required. The iGPU can suffer an unrecoverable MES (Micro Engine Scheduler) hang during sustained inference without amdgpu.noretry=0. Do not skip.

```
sudo rpm-ostree kargs --append=amdgpu.noretry=0
systemctl reboot
```

After reboot, verify the parameter is active before proceeding:

```
cat /proc/cmdline | grep amdgpu.noretry
→ output must contain "amdgpu.noretry=0"
```

Required Environment Variables (in docker-compose.yml)

Both must be set in the Ollama container's environment block. Setting them in the host shell has no effect on containers.

Variable	Value	Effect
HSA_OVERRIDE_GFX_VERSION	11.0.2	Bypasses ROCm hardware allowlist. Without it, Ollama silently falls back to CPU.
OLLAMA_FLASH_ATTENTION	false	RDNA3 has a numerical instability bug in Flash Attention causing output to collapse into repeated tokens (e.g. "444444..."). Disabling it fixes this with negligible performance impact.

BIOS Prerequisite

The 780M uses system RAM as VRAM. A 14B Q4 model requires approximately 9 GB. The BIOS UMA Frame Buffer Size must be set to "Auto" or ≥ 8 GB. GMKtec NucBox K8 default: Auto. Verify in BIOS → APU/Advanced → UMA Frame Buffer Size.

Expected Performance

Mode	Model	Speed
CPU-only	7B Q4	~3–6 tok/s
780M GPU (recommended)	7B Q4	~15–20 tok/s
780M GPU	14B Q4	~10–14 tok/s
Voice end-to-end latency	CPU fallback	~30–40 s
Voice end-to-end latency	780M GPU	~8–12 s

> **INFO** eGPU note: when an OCuLink eGPU (RX 6000/7000 series) is added, Ollama auto-selects it with no configuration changes. Use ROCR_VISIBLE_DEVICES if explicit device selection is needed.

SYSTEM ARCHITECTURE

Components

Service	Image	Folder
Ollama (GPU)	ollama/ollama:rocm	~/docker/ollama
Open WebUI	ghcr.io/open-webui/open-webui:main	~/docker/openwebui
mcpo (MCP proxy)	ghcr.io/open-webui/mcpo:main	~/docker/mcpo

Service	Image	Folder
Home Assistant	ghcr.io/home-assistant/home-assistant:stable	~/docker/homeassistant
wyoming-whisper (STT)	rhasspy/wyoming-whisper:latest	~/docker/wyoming
wyoming-piper (TTS)	rhasspy/wyoming-piper:latest	~/docker/wyoming
Mosquitto (future)	eclipse-mosquitto:latest	~/docker/zigbee
Zigbee2MQTT (future)	ghcr.io/koenkk/zigbee2mqtt:latest	~/docker/zigbee

Docker Network Topology

CONCEPT Create the ai-net bridge network once before deploying any AI stack container. It is a persistent host resource — it survives compose up/down cycles.
docker network create ai-net

Network	Members	Reason
ai-net (bridge)	ollama, openwebui, mcpo	Containers communicate by hostname (e.g. http://ollama:11434)
host network	homeassistant	Required for mDNS/UPnP device discovery. HA cannot join bridge networks.
default (published ports)	wyoming-whisper, wyoming-piper	Ports bound to 127.0.0.1 — HA reaches them via loopback as host network shares it

CONCEPT Hostname resolution summary:
openwebui → ollama:11434 (ai-net internal Docker hostname)
openwebui → mcpo:8000 (ai-net internal Docker hostname)
homeassistant → 127.0.0.1:11434 (loopback — HA is on host network)
homeassistant → 127.0.0.1:10300 (wyoming-whisper)
homeassistant → 127.0.0.1:10200 (wyoming-piper)
NPM → <LAN-IP>:3000 and <LAN-IP>:8123 (cross-network: use LAN IP, not container name)

Port Registry

Port	Binding	Service	External Access
22022	0.0.0.0	SSH	Yes (existing router forward)
80 / 443	0.0.0.0	NPM HTTP/HTTPS	Yes (existing router forward)
11434	127.0.0.1	Ollama API	Loopback only — LAN devices cannot reach it
3000	0.0.0.0	Open WebUI	Via NPM → ai.*.duckdns.org (LAN also has plain HTTP — see Note)
8000	ai-net only	mcpo	Docker internal — not published to host
8123	host network	Home Assistant	Via NPM → home.*.duckdns.org
10300	127.0.0.1	wyoming-whisper	Loopback only
10200	127.0.0.1	wyoming-piper	Loopback only
1883 (future)	127.0.0.1	Mosquitto MQTT	Loopback only
8282 (future)	127.0.0.1	Zigbee2MQTT UI	Loopback only

NOTE Open WebUI binds to 0.0.0.0:3000 because NPM (on nextcloud-aio network) must reach it via the server's LAN IP — cross-network container hostname resolution does not work between nextcloud-aio and ai-net without adding ai-net to NPM. As a result, http://<LAN-IP>:3000 is accessible unencrypted on your LAN. Open WebUI has its own authentication; all external traffic goes through HTTPS via NPM. For a trusted home LAN this is the standard accepted trade-off.
No new router port forwards are required. All external access uses the existing NPM rules.

Data Flow

```
Browser / Android / Linux laptop
  ↓ HTTPS via DuckDNS + NPM (existing, unchanged)
Open WebUI (ai-net)  ↔  mcpo (ai-net)  ↔  MCP servers (filesystem, memory)
  ↓ http://ollama:11434 (ai-net internal)
Ollama :rocm  [780M iGPU · qwen2.5:14b-instruct-q4_K_M]
  ↑ http://127.0.0.1:11434 (loopback)
Home Assistant (host network)
  ⇔ 127.0.0.1:10300  wyoming-whisper (STT)
  ⇔ 127.0.0.1:10200  wyoming-piper (TTS)
  ↓ ESPHome voice_assistant over Wi-Fi (future)
ESP32 satellite nodes
```

COMPONENT RATIONALE & FULL COMPOSE BLOCKS

Ollama (ollama/ollama:rocm)

The :rocm tag is mandatory. The untagged ollama/ollama image is CPU-only and silently ignores all GPU environment variables.

```
services:
  ollama:
    image: ollama/ollama:rocm
    container_name: ollama
    restart: unless-stopped
    labels:
      - "com.centurylinklabs.watchtower.enable=true"
    ports:
      - "127.0.0.1:11434:11434"
    devices:
      - /dev/kfd:/dev/kfd
      - /dev/dri:/dev/dri
    group_add:
      - "<VIDEO_GID>"         # numeric GID from EX.0.d — remove angle brackets, e.g. "44"
      - "<RENDER_GID>"        # numeric GID from EX.0.d — remove angle brackets, e.g. "107"
    environment:
      - HSA_OVERRIDE_GFX_VERSION=11.0.2
      - OLLAMA_FLASH_ATTENTION=false
      - OLLAMA_KEEP_ALIVE=5m
      - OLLAMA_NUM_PARALLEL=1
      - OLLAMA_HOST=0.0.0.0
    volumes:
      - ./models:/root/.ollama
    networks:
      - ai-net

networks:
  ai-net:
    external: true
```

Variable	Value	Reason
HSA_OVERRIDE_GFX_VERSION	11.0.2	Forces ROCm to treat gfx1103 as a supported device
OLLAMA_FLASH_ATTENTION	false	Disables RDNA3 Flash Attention bug — mandatory for coherent output
OLLAMA_KEEP_ALIVE	5m	Model stays loaded 5 minutes after last request. Adjust lower if RAM pressure is observed.
OLLAMA_NUM_PARALLEL	1	One concurrent request. Prevents VRAM exhaustion on shared-memory iGPU.
OLLAMA_HOST	0.0.0.0	Ollama listens on all container interfaces — required so OpenWebUI can reach it by container hostname on ai-net.

CONCEPT Why numeric GIDs in group_add? Docker resolves string group names from the container's own /etc/group, not the host's. If the group is absent from the container image, startup crashes with "Unable to find group render". Obtain numeric GIDs with: getent group video | cut -d: -f3 and getent group render | cut -d: -f3

CONCEPT Recommended model: qwen2.5:14b-instruct-q4_K_M — valid Ollama tag, 9.0 GB, 10–14 tok/s on 780M.
Fallback: mistral:7b-instruct-q4_K_M (~5 GB) if available RAM drops below 18 GB.

Open WebUI (ghcr.io/open-webui/open-webui:main)

```
services:
  openwebui:
    image: ghcr.io/open-webui/open-webui:main
    container_name: openwebui
    restart: unless-stopped
    labels:
      - "com.centurylinklabs.watchtower.enable=true"
    ports:
      - "3000:8080"
    environment:
      - OLLAMA_BASE_URL=http://ollama:11434
      - WEBUI_SECRET_KEY=<paste output of: openssl rand -hex 32>
    volumes:
      - ./data:/app/backend/data
    networks:
      - ai-net

networks:
  ai-net:
    external: true
```

NOTE WEBUI_SECRET_KEY: run "openssl rand -hex 32" in the terminal first and paste the resulting 64-character hex string. Do not use the placeholder literally. Remove the angle brackets.

mcpo (ghcr.io/open-webui/mcpo:main)

mcpo is a Python server that proxies MCP (Model Context Protocol) servers over HTTP/OpenAPI, making them accessible to Open WebUI as tool servers. It runs npx to launch the Node.js-based MCP server packages — the mcpo Docker image includes Node.js for this purpose.

CRITICAL mcpo requires an explicit --config /app/config/config.json startup command. Without it the container starts with no MCP servers loaded and exposes no tools.

config.json

```
{
  "mcpServers": {
    "filesystem": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem", "/nc-data"]
    },
    "memory": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-memory"],
      "env": {
        "MEMORY_FILE_PATH": "/app/memory/graph.json"
      }
    }
  }
}
```

CONCEPT MEMORY_FILE_PATH is an environment variable read by the memory server package, not a CLI argument. Without it the memory graph is not persisted to disk and is lost on every container restart.

docker-compose.yml

```
services:
  mcpo:
    image: ghcr.io/open-webui/mcpo:main
    container_name: mcpo
    restart: unless-stopped
    command: --config /app/config/config.json
    labels:
      - "com.centurylinklabs.watchtower.enable=true"
    user: "<NC_UID>:<NC_GID>"    # from EX.0.g — remove angle brackets, e.g. "33:33"
    environment:
      - npm_config_cache=/tmp/npm_cache
      - HOME=/tmp
    volumes:
      - ./config.json:/app/config/config.json:ro
      - /var/srv/nextcloud-data/<your_nc_username>/files:/nc-data:ro
      - ./memory:/app/memory
    networks:
      - ai-net

networks:
  ai-net:
    external: true
```

Setting	Reason
user: "33:33"	Runs mcpo as the Nextcloud data owner so it can read user files. Remove angle brackets — type the numeric values directly.
npm_config_cache=/tmp	Prevents EACCES: npx writes cache to the user home directory, which does not exist for a non-root UID without a /etc/passwd entry.
HOME=/tmp	Node/npm resolves HOME from /etc/passwd. An arbitrary UID has no entry, causing EACCES crashes. Setting HOME=/tmp provides a writable virtual home directory. Mandatory.
/var/srv/nextcloud-data/<user>/files:/nc-data:ro	Scopes AI filesystem access to one user's files directory. The full data path includes system folders (appdata_*, updater-*) that pollute the AI context. :ro prevents any AI write to Nextcloud.
./memory:/app/memory	Persists the memory graph.json across container restarts. Must be pre-created with correct ownership before first start.

CRITICAL The ./memory directory must be pre-created with correct ownership BEFORE docker compose up. Docker auto-creates missing bind-mount directories as root:root. mcpo runs as a non-root UID and cannot write the memory graph, causing silent failure.
mkdir -p ~/docker/mcpo/memory
sudo chown <NC_UID>:<NC_GID> ~/docker/mcpo/memory # use numeric values, no angle brackets

CONCEPT Connecting Open WebUI to mcpo requires registering each MCP server at its individual path. The root URL (http://mcpo:8000) exposes no tool definitions.
Admin Settings → Tools → "+" (Add Tool Server)
URL: http://mcpo:8000/filesystem Type: OpenAPI Auth: None
URL: http://mcpo:8000/memory Type: OpenAPI Auth: None
Enable both tools globally in Admin Settings and per model in model config.

WARNING Nextcloud Server-Side Encryption (SSE): if SSE is enabled on your Nextcloud instance, files inside nextcloud-data/<user>/files/ are encrypted blobs. The AI reads only unreadable binary data regardless of path. SSE must be disabled for this user. Verify: Nextcloud web UI → Settings → Administration → Security → "Server-side encryption" must be off.

Home Assistant (ghcr.io/home-assistant/home-assistant:stable)

```
services:
  homeassistant:
    image: ghcr.io/home-assistant/home-assistant:stable
    container_name: homeassistant
    restart: unless-stopped
    network_mode: host
    labels:
      com.centurylinklabs.watchtower.enable: "false"
    environment:
      - TZ=Europe/Berlin          # adjust to your timezone
    volumes:
      - ./config:/config
      - /etc/localtime:/etc/localtime:ro
      - /run/dbus:/run/dbus:ro
```

Setting	Reason
network_mode: host	Required for mDNS/UPnP device discovery. HA cannot join bridge networks. No ports: block — that is incompatible with host mode.
privileged: true (absent)	NOT required for HA Container. It is required by HA Supervisor but grants unrestricted host root access — inappropriate for an internet-facing service.
watchtower.enable: "false"	Prevents Watchtower from auto-restarting HA mid-session. Update HA manually through docker compose pull.
/run/dbus:ro	Optional for this core build. Required for future Bluetooth device tracking and system-level integrations.

CRITICAL HA add-ons do not exist in Container mode. Do not follow any tutorial referencing the Supervisor, Add-on Store, or named add-ons — those apply to HAOS or Supervised installations only.

CONCEPT Reverse proxy configuration (required before remote access): since HA 2021.7, reverse proxy requests are rejected by default with "400 Bad Request". Edit ~/docker/homeassistant/config/configuration.yaml after onboarding completes:
http:
  use_x_forwarded_for: true

```
trusted_proxies:
  - 172.16.0.0/12    # standard Docker bridge range
  - 192.168.0.0/16   # guard for non-standard Docker allocations
  - 127.0.0.1        # loopback
```
Validate before restarting — a YAML indentation error prevents HA from booting.

Voice Pipeline: wyoming-whisper + wyoming-piper (~/docker/wyoming)

Both services in one docker-compose.yml. Ports bound to 127.0.0.1 — Home Assistant (host network) reaches them via the shared loopback. LAN devices cannot reach these ports directly.

```
services:
  wyoming-whisper:
    image: rhasspy/wyoming-whisper:latest
    container_name: wyoming-whisper
    restart: unless-stopped
    labels:
      - "com.centurylinklabs.watchtower.enable=true"
    ports:
      - "127.0.0.1:10300:10300"
    volumes:
      - ./whisper-data:/data
    command: --model base --language en

  wyoming-piper:
    image: rhasspy/wyoming-piper:latest
    container_name: wyoming-piper
    restart: unless-stopped
    labels:
      - "com.centurylinklabs.watchtower.enable=true"
    ports:
      - "127.0.0.1:10200:10200"
    volumes:
      - ./piper-data:/data
    command: --voice en_US-lessac-high
```

NOTE Whisper model options: tiny (fastest, less accurate) · base (good balance for home commands) · small · medium (best accuracy, slower). Change --model base to --model small for better recognition of accented speech.
In HA: Settings → Devices & Services → Add Integration → Wyoming Protocol. Add twice: 127.0.0.1:10300 → Whisper STT, 127.0.0.1:10200 → Piper TTS.

NPM Proxy Requirements (both proxy hosts)

NPM cannot reach Open WebUI or Home Assistant by Docker container name — they are on different networks. Use the server's LAN IP for both forward targets.

Proxy Host	Forward Host	Port	Required Settings
ai.yourdomain.duckdns.org	<LAN-IP>	3000	WebSockets ON · SSL ON · Cache Assets ON · Block Exploits ON
home.yourdomain.duckdns.org	<LAN-IP>	8123	WebSockets ON · SSL ON · Cache Assets ON · Block Exploits ON

WARNING WebSockets Support must be ON for both proxy hosts. Open WebUI streaming and the HA event bus both require WebSockets — turning it off breaks real-time updates and AI response streaming.

Ollama Conversation Agent in Home Assistant

Because Home Assistant runs on the host network, it cannot resolve Docker container hostnames. Use the loopback address 127.0.0.1 — not the container name "ollama".

```
Settings → Devices & Services → Add Integration → Ollama
  URL:    http://127.0.0.1:11434
  Model:  qwen2.5:14b-instruct-q4_K_M
  Enable "Control Home Assistant"
  Context window: 8192
```

IMPLEMENTATION PHASES

Phase 0 Kernel Parameter — Mandatory · Requires Reboot

CRITICAL This is the first step. Do not start any other phase until this is confirmed active. Without amdgpu.noretry=0 the iGPU can suffer an unrecoverable MES hang during sustained inference, requiring a hard reboot.

```
sudo rpm-ostree kargs --append=amdgpu.noretry=0
systemctl reboot
```

After reboot, verify before continuing:

```
cat /proc/cmdline | grep amdgpu.noretry
→ must contain "amdgpu.noretry=0"
```

EX.0 Pre-flight (~20 min)

All values gathered here feed directly into compose files. Record each output.

Step a — Verify existing containers healthy

```
docker ps --format "table {{.Names}}\t{{.Status}}"
→ duckdns, npm, nextcloud-aio-mastercontainer: all "Up"
```

Step b — Confirm kernel parameter (Phase 0 must already be done)

```
cat /proc/cmdline | grep amdgpu.noretry
→ must contain "amdgpu.noretry=0"
```

Step c — Verify GPU device nodes exist

```
ls -la /dev/kfd /dev/dri/renderD128
→ both files must be present
```

Step d — Retrieve numeric GIDs for Ollama group_add — record both numbers

```
getent group video  | cut -d: -f3   → e.g. 44
getent group render | cut -d: -f3   → e.g. 107
```

CONCEPT cut -d: -f3 extracts only the third colon-separated field (the numeric GID). Record both numbers — you will paste them directly into the Ollama compose file.

Step e — Verify available RAM

```
free -h | awk '/^Mem:/{print $7}'
→ "Available" (includes reclaimable cache). Aim for ≥ 18 GB before starting Ollama.
```

Step f — Verify BIOS UMA Frame Buffer Size

Boot into BIOS → APU/Advanced → UMA Frame Buffer Size. Must be "Auto" or ≥ 8 GB. GMKtec NucBox K8 default: Auto. If set to a small fixed value, change it to Auto and save.

Step g — Retrieve Nextcloud data directory owner UID:GID — record both numbers

```
ls -lnd /var/srv/nextcloud-data
# -n: forces numeric UID/GID output
# -d: lists the directory itself, not its contents
→ Note the third (UID) and fourth (GID) fields
→ Example: drwxrwx--- 1 33 33 ... /var/srv/nextcloud-data
#   UID=33, GID=33
```

Step h — Identify your Nextcloud username

```
ls /var/srv/nextcloud-data/
→ your username appears as a directory alongside appdata_* and others
```

Step i — Verify Nextcloud SSE is disabled

Nextcloud web UI: Settings → Administration → Security → "Server-side encryption" must be off. If SSE is enabled, the filesystem tool returns only unreadable encrypted blobs.

Step j — Create shared Docker network

```
docker network create ai-net
# If "network with name ai-net already exists" appears, that is correct — no action needed.
```

EX.1 Ollama with iGPU (~45 min — mostly model download)

```
mkdir -p ~/docker/ollama && cd ~/docker/ollama
nano docker-compose.yml
```

Write the Ollama compose block from the Component Rationale section. Substitute:

- **VIDEO_GID** — the numeric GID from EX.0.d (e.g. 44). Remove angle brackets.
- **RENDER_GID** — the numeric GID from EX.0.d (e.g. 107). Remove angle brackets.

```
docker compose up -d
```

Pull the recommended model (~9.0 GB, one-time download):

```
docker exec ollama ollama pull qwen2.5:14b-instruct-q4_K_M
```

Verify GPU offload (run during an active generation):

```
docker exec ollama ollama ps
→ PROCESSOR column must show "100% GPU"
# If it shows CPU: confirm HSA_OVERRIDE_GFX_VERSION is set:
docker inspect ollama | grep -A1 HSA_OVERRIDE
```

Functional test:

```
curl http://127.0.0.1:11434/api/generate \
  -d '{"model":"qwen2.5:14b-instruct-q4_K_M","prompt":"Hello","stream":false}'
→ JSON with a "response" field.
```

> **NOTE** First-request latency: the initial request after starting Ollama takes up to 30 seconds while the model loads into VRAM. This is normal. All subsequent requests will be fast (~8–12 seconds end-to-end for a short reply at 780M speeds).

LAN isolation test:

```
# From another device on your LAN:
curl http://<server-LAN-IP>:11434
→ Connection refused
```

EX.2 Open WebUI + mcpo (~45 min)

Open WebUI

```
mkdir -p ~/docker/openwebui && cd ~/docker/openwebui

# Generate a secret key — copy the output:
openssl rand -hex 32

nano docker-compose.yml
# Write the Open WebUI compose block from Component Rationale.
# Paste the generated key as WEBUI_SECRET_KEY value. Remove angle brackets.

docker compose up -d
```

mcpo

```
mkdir -p ~/docker/mcpo && cd ~/docker/mcpo
```

Write config.json:

```
nano config.json
# Write the config.json block from Component Rationale.
# The Nextcloud filesystem volume path uses your actual username from EX.0.h.
```

Pre-create the memory directory with correct ownership (mandatory before compose up):

```
# Replace <NC_UID> and <NC_GID> with the numeric values from EX.0.g.
# Remove angle brackets — type the numbers directly, e.g. "33:33".
mkdir -p ~/docker/mcpo/memory
sudo chown <NC_UID>:<NC_GID> ~/docker/mcpo/memory
```

Write docker-compose.yml:

```
nano docker-compose.yml
# Write the mcpo compose block from Component Rationale.
# Substitute NC_UID:NC_GID (from EX.0.g) and nc_username (from EX.0.h).
# Remove all angle brackets — type numeric values and username directly.

docker compose up -d
```

Verify mcpo started successfully:

```
docker logs mcpo | head -40
→ "Successfully connected to: filesystem" must appear
```

```
→ "Successfully connected to: memory" must appear
→ "INFO: Application startup complete." must appear

# Verify Node.js is available (should pass with the official mcpo image):
docker exec mcpo node --version && docker exec mcpo npx --version
→ both must return version strings, not "executable not found"
```

Connect Open WebUI to mcpo

```
Open WebUI UI → Admin Settings → Tools → "+" (Add Tool Server)
  URL: http://mcpo:8000/filesystem   Type: OpenAPI   Auth: None → Save
  URL: http://mcpo:8000/memory       Type: OpenAPI   Auth: None → Save
Enable both tools globally in Admin Settings.
Per model: Model config → enable both tools.
```

Add NPM proxy host for Open WebUI

```
NPM → Hosts → Proxy Hosts → Add Proxy Host
  Domain: ai.yourdomain.duckdns.org
  Forward Hostname: <LAN-IP>   Port: 3000
  WebSockets Support: ON   SSL: Let's Encrypt   Force SSL: ON
```

Functional tests

```
1. "List the files in /nc-data" → AI returns your Nextcloud files
2. "Remember that my test value is 42" → AI confirms storage
3. "What is my test value?" → returns 42
4. docker compose restart mcpo
   "What is my test value?" → still returns 42 (confirms persistence)
```

EX.3 Home Assistant (~60 min)

```
mkdir -p ~/docker/homeassistant && cd ~/docker/homeassistant
nano docker-compose.yml
# Write the Home Assistant compose block from Component Rationale.
# No ports: block — network_mode: host is incompatible with it.

docker compose up -d
```

Step 1 — Complete HA onboarding first

CRITICAL Do NOT edit configuration.yaml yet. HA generates this file during onboarding — editing before that step creates a conflicting empty file.
Open http://<LAN-IP>:8123 and complete the full onboarding wizard.
Wait until the HA dashboard is fully visible before proceeding to the next step.

Step 2 — Configure reverse proxy trust

```
# Only after onboarding completes and the dashboard is visible:
nano ~/docker/homeassistant/config/configuration.yaml

# Add (or insert into existing http: block if present):
http:
  use_x_forwarded_for: true
  trusted_proxies:
    - 172.16.0.0/12    # standard Docker bridge range (172.x.x.x)
    - 192.168.0.0/16   # guard for non-standard Docker allocations
    - 127.0.0.1        # loopback
```

Validate the YAML before restarting — a single indentation error prevents HA from booting:

```
docker exec homeassistant python -m homeassistant \
  --script check_config --config /config
→ must print "Configuration is valid!"
# If error: fix YAML indentation before proceeding.
```

Restart only after validation passes:

```
cd ~/docker/homeassistant && docker compose restart
```

Step 3 — Add NPM proxy host for HA

```
NPM → Hosts → Proxy Hosts → Add Proxy Host
  Domain: home.yourdomain.duckdns.org
  Forward Hostname: <LAN-IP>   Port: 8123
  WebSockets Support: ON   SSL: Let's Encrypt   Force SSL: ON

In HA: Settings → System → Network → External URL →
  https://home.yourdomain.duckdns.org
```

Step 4 — Verify remote access

```
Navigate to https://home.yourdomain.duckdns.org
→ HA dashboard loads with no "Unable to connect" error toast

# If 400 Bad Request: check docker logs nginx-proxy-manager for the source IP.
# Confirm that trusted_proxies range in configuration.yaml covers it.
```

EX.4 Voice Pipeline (~30 min)

```
mkdir -p ~/docker/wyoming && cd ~/docker/wyoming
nano docker-compose.yml
# Write the Wyoming compose block from Component Rationale.

docker compose up -d
```

Add Wyoming integrations in HA:

```
Settings → Devices & Services → Add Integration → Wyoming Protocol
  Add twice using 127.0.0.1:
    127.0.0.1 : 10300  →  Faster Whisper (STT)
    127.0.0.1 : 10200  →  Piper (TTS)
```

Create the Assist pipeline:

```
Settings → Voice Assistants → Add Assistant
  Name: Wall-E
  STT: Faster Whisper
  TTS: Piper
  (Conversation Agent: set in EX.5)
```

STT test:

```
HA Assist panel → microphone icon → speak a short sentence.
→ Verify the transcript appears before proceeding to EX.5.
```

EX.5 HA → Ollama Conversation Agent (~20 min)

```
Settings → Devices & Services → Add Integration → Ollama
  URL:    http://127.0.0.1:11434
  Model:  qwen2.5:14b-instruct-q4_K_M
  Enable "Control Home Assistant"
  Context window: 8192
```

CONCEPT HA uses 127.0.0.1 because it is on host network and cannot resolve the container hostname "ollama". Only ai-net bridge containers can do that.

Assign to the Wall-E Assist pipeline:

```
Settings → Voice Assistants → Wall-E
  Conversation Agent → select the Ollama integration
```

Optional system prompt:

```
"You are Wall-E, a helpful home assistant. Answer concisely.
 Report device states accurately when asked."
```

End-to-end tests:

```
# Typed test:
Settings → Voice Assistants → Wall-E → three-dot → Start conversation
Type: "What time is it?"
→ Ollama responds, Piper speaks. First query ~15 s. Subsequent: ~8–12 s.

# Voice test:
Browser mic in HA Assist. Speak a command.
→ Verify: transcript visible → AI response → voice output.
```

EX.6 Remote Access & Mobile Clients (~20 min)

Verify HTTPS from browser or phone:

```
https://ai.yourdomain.duckdns.org     → Open WebUI loads
https://home.yourdomain.duckdns.org   → HA loads without errors
```

Install HA Companion App (Android):

```
1. Google Play → "Home Assistant" → install
2. Login with: https://home.yourdomain.duckdns.org
3. Settings → Companion App → Voice → enable microWakeWord
4. Test: say wake word → command → AI responds via phone speaker
```

NOTE Linux laptop: open both HTTPS URLs in any browser. No further setup required.
Open WebUI can be added to the Android home screen via Chrome → "Add to Home Screen" for a PWA-style install. Allow microphone permission for voice input.

CONCEPT —— CORE BUILD COMPLETE ——
At the end of EX.6:
√ AI chat via browser on any device, anywhere over HTTPS
√ Voice via Android Companion App (hands-free, on-phone wake word)
√ Persistent memory (store, recall, delete — survives restarts)
√ Nextcloud file access (plain text, PDFs, config files)
√ HA voice assistant with Ollama conversation agent
√ HA device control via AI ("turn on the living room light")

> ✓ All main guide containers (Nextcloud, DuckDNS, NPM) unchanged

POST-EXECUTION VALIDATION CHECKLIST

Run in order. Each check must pass before the next is meaningful.

#	Check	Expected Result
1	Kernel parameter active: `cat /proc/cmdline | grep amdgpu.noretry`	"amdgpu.noretry=0" present in output
2	ai-net exists: `docker network ls | grep ai-net`	One line returned showing ai-net with driver "bridge"
3	GPU inference confirmed (run during active generation): `docker exec ollama ollama ps`	PROCESSOR column reads "100% GPU"
4	LAN isolation verified (from another LAN device): `curl http://<server-LAN-IP>:11434`	Connection refused
5	mcpo memory directory ownership: `stat -c "%u:%g" ~/docker/mcpo/memory`	Matches NC_UID:NC_GID from EX.0.g
6	mcpo memory persistence: "Remember test word Alpha" → restart mcpo → "What test word?"	Returns "Alpha"
7	HA config validated: `docker exec homeassistant python -m homeassistant --script check_config --config /config`	"Configuration is valid!"
8	HA WebSocket integrity: https://home.yourdomain.duckdns.org → dashboard	Loads with no error toast
9	Voice pipeline end-to-end: HA Assist or Android: "What is today's date?"	STT transcribes → Ollama responds → Piper speaks. All local.
10	Remote access from mobile data (not home Wi-Fi): https://ai.yourdomain.duckdns.org and https://home.yourdomain.duckdns.org	Both load correctly

CRITICAL PITFALLS

Ordered by phase. Each pitfall has caused real deployment failures.

1. Kernel parameter before everything.

rpm-ostree kargs + reboot required first. iGPU hangs the OS under sustained inference without amdgpu.noretry=0.

2. Wrong Ollama image tag.

ollama/ollama:rocm is the GPU image. The untagged image is CPU-only and silently ignores all GPU environment variables.

3. HSA_OVERRIDE_GFX_VERSION must be in the compose environment block.

Shell-level environment variables do not propagate into containers.

4. OLLAMA_FLASH_ATTENTION=false is mandatory on gfx1103.

RDNA3 Flash Attention causes output to collapse into repeated tokens (e.g. "444444...")

5. **Numeric GIDs in group_add, not string names.**

   String names are resolved from the container's /etc/group, not the host's. Startup crashes if the group is absent from the container image.

6. **ai-net must exist before docker compose up on any stack that references it.**

   Run "docker network create ai-net" (EX.0.j) before all other steps.

7. **Ollama and Wyoming ports bound to 127.0.0.1, not 0.0.0.0.**

   HA connects via 127.0.0.1 (loopback, host network). Setting HA's Ollama URL as "http://ollama:11434" will fail — HA is not on ai-net and cannot resolve that hostname.

8. **mcpo requires command: --config /app/config/config.json.**

   Without it the container starts with no tools loaded.

9. **MEMORY_FILE_PATH is an env var inside config.json, not a CLI argument.**

   Without it, the memory graph is not persisted across restarts.

10. **mcpo tools must be registered individually in Open WebUI.**

    http://mcpo:8000/filesystem and http://mcpo:8000/memory — not the root URL. The root URL exposes no tool definitions.

11. **mcpo memory directory must be pre-chowned before docker compose up.**

    Docker auto-creates missing bind-mount directories as root:root. sudo chown <NC_UID>:<NC_GID> ~/docker/mcpo/memory before compose up.

12. **npm_config_cache=/tmp/npm_cache in mcpo environment.**

    npx running as a non-root UID fails with EACCES on the default cache path.

13. **HOME=/tmp in mcpo environment is mandatory.**

    Running as an arbitrary UID (e.g. 33:33) provides no /etc/passwd entry inside the container. Node and npm resolve HOME from /etc/passwd — without it, startup crashes with EACCES. HOME=/tmp provides a writable virtual home directory.

14. **ls -lnd for Nextcloud UID:GID — both flags are mandatory.**

    -n forces numeric output; -d lists the directory itself, not its contents. Omitting either flag gives wrong or ambiguous results.

15. **mcpo filesystem volume: mount user files only.**

    /var/srv/nextcloud-data/<username>/files:/nc-data:ro — note "nextcloud-data", not "nextcloud/data". The full data directory includes system folders that pollute the AI context.

16. **Nextcloud SSE must be disabled.**

    If server-side encryption is enabled, files/ contains encrypted blobs. The AI reads unreadable binary data regardless of path scoping.

17. **HA configuration.yaml trusted_proxies required before remote access.**

    Without it: 400 Bad Request. Validate YAML before restarting HA.

18. **Complete HA onboarding before editing configuration.yaml.**

    HA generates configuration.yaml during onboarding. Editing or creating it before that step causes a conflict and prevents HA from booting.

19. **NPM WebSockets Support must be ON for both proxy hosts.**

    Open WebUI streaming and HA event bus both require WebSockets.

20. **HA network_mode: host — no ports: block, no privileged: true.**

    privileged: true is not required for HA Container. It grants unrestricted host access and is inappropriate for an internet-facing service. For future Zigbee: use targeted devices: entry, not privileged mode.

21. **HA add-ons do not exist in Container mode.**

    Do not follow tutorials referencing Supervisor, Add-on Store, or add-ons.

22. **Watchtower disabled on HA.**

    com.centurylinklabs.watchtower.enable: "false" prevents mid-session restarts. Update HA manually.

23. **NPM forward targets use the server's LAN IP.**

    Cross-network container hostname resolution does not work between nextcloud-aio and ai-net. Never use a container name like "openwebui" in NPM's forward host field — use the server's LAN IP.

24. **Remove angle brackets from all placeholders before running.**

    e.g. user: "33:33" NOT user: "<33>:<33>". Angle brackets in a YAML value cause a Docker Compose deployment error.

HARDWARE APPENDIX — Future Phases

INFO EX.0 through EX.6: no new hardware required. All items below are for future extension only.

ESP32 Voice Satellite Nodes

Existing ESP32 WROOM boards are usable. Wake word runs server-side in HA via openWakeWord. ESPHome firmware with voice_assistant component.

Component	Purpose	Cost
INMP441 I2S microphone breakout	Microphone input	~€3 each
MAX98357A I2S amplifier breakout	Audio output	~€3 each
Small 4 Ω passive speaker	Speaker output	~€5 each
USB-C power (any phone charger)	Power supply	€0
ESP32-S3 N8R8 (optional upgrade)	On-device wake word (replaces WROOM board only, same wiring)	~€8 each

Zigbee / Thread Hub

Option	Protocol	Cost
HA Connect ZBT-2 (recommended)	Zigbee + Thread (Matter)	~€30
Sonoff ZBDongle-E	Zigbee only	~€20
USB 2.0 extension cable 20–30 cm	Always required — USB RF interference from CPU	~€3

NOTE When adding a Zigbee dongle to the HA compose file, use a targeted devices: entry pointing to /dev/serial/by-id/<dongle-id>. Never use privileged: true — it is not needed and grants unrestricted host access.